

INDIAN RAILWAYS · VANDE BHARAT

VandInspect AI



How the Automated Train Inspection System Works

A plain-language guide — no technical background needed

Prepared for non-technical readers · 2026

Includes: System Architecture · The OCR Speed Trick (ROI) · The Bogie Image Allocation System

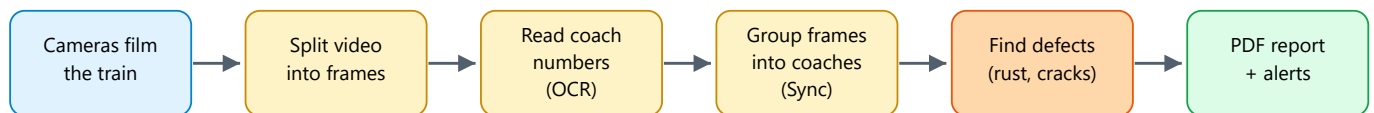
1. The Big Picture — What This System Does

Imagine a train slowly rolling through a special inspection shed. Cameras line both sides and the floor, filming every coach as it passes. Today, a human engineer would walk the length of the train with a torch — for a 16-coach Vande Bharat that takes **60–90 minutes** and tired eyes miss small cracks.

VandelInspect AI replaces that torch-and-clipboard process. It watches the same video, automatically finds every coach, reads each coach's printed number, hunts for defects like rust and cracks, and produces a ready-to-sign inspection report — in a fraction of the time and without human fatigue.

In one sentence: Videos go in → the system splits the train into coaches, reads their numbers, spots damage, and prints an audit-ready PDF report.

The journey of a video, end to end

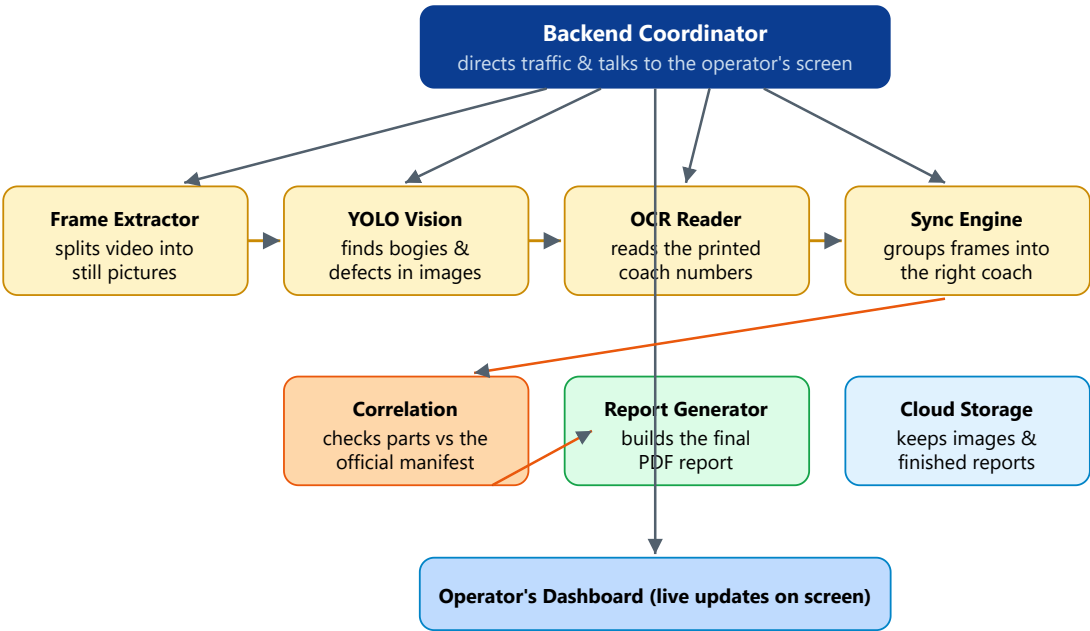


Each box is an automatic step. A human only steps in at the very end to review and sign.

Why it matters: faster turnaround between journeys, no missed defects from human fatigue, and every defect is automatically tied to the exact coach number — no more guessing from paper logs.

2. System Architecture — The Team of Specialists

Rather than one giant program doing everything, the system is built as a **team of small independent services**, each an expert at one job. They hand work to each other like an assembly line. If one is busy or restarts, the others are unaffected.



Eight cooperating services. Work flows left-to-right; the coordinator oversees everyone.

Specialist	Its single job, in plain words
Frame Extractor	Chops the uploaded videos into thousands of still pictures and labels each one with a sequence number.
YOLO Vision	An "eyes" model. In one picture it can point to the bogie (the wheel-and-frame unit under a coach) and separately spot defects like cracks or rust.
OCR Reader	Reads the actual <i>printed digits</i> of a coach number off the metal — like a person reading a number plate.
Sync Engine	Decides which pictures belong to which coach, and lines up all the cameras to the same moment in time.
Correlation	Compares the parts it sees against the official "this coach should have these parts" list and flags anything missing.
Report Generator	Packages everything into a clean PDF an engineer can sign off.
Cloud Storage + Dashboard	Stores images/reports and shows live progress on the operator's screen.

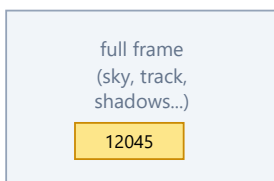
3. The OCR Speed Trick — "Look Only Where It Matters" (ROI)

Reading text off an image (OCR) is slow and expensive. A single camera frame is huge and is mostly sky, track, ballast and shadows — the coach number occupies a tiny patch. Asking the reader to scan the *entire* picture wastes time and invites mistakes (it might "read" numbers in dirt or rivets).

The core idea — ROI (Region of Interest): First let the fast vision model draw a box around just the bogie/number area, then crop the picture down to that box, and only run the slow reader on that small crop.

Analogy: instead of reading every page of a newspaper to find one phone number, a helper first circles the box where the number is — you only read inside the circle.

The slow, old way



scan EVERYTHING →
slow + false reads

The ROI way (this system)



Left: reading the whole frame is slow and error-prone. Right: crop to the boxed area first, then read.

How it works, step by step

- **Step 1 — Spot the bogie.** The fast YOLO Vision model looks at the frame and returns a box around the bogie / number region (it is trained to recognise parts like **Bogie** **Engine** **Car Type**).
- **Step 2 — Crop with a small margin.** The system cuts out that box plus a **15% padding** on each side, so a slightly-off box still keeps the full number inside.
- **Step 3 — Read the small crop.** The OCR reader runs on just that crop — far less pixels, far faster, far fewer false readings.
- **Step 4 — Try harder only if needed.** If the first read is blank, it cleans up the crop (sharpening and contrast) and reads again. Only as a last resort does it fall back to reading the full frame.
- **Step 5 — Trust but verify.** A result is only accepted if it looks like a real coach number (5–6 digits) with high enough confidence.

The payoff: dramatically faster OCR and much higher accuracy, because the reader spends its effort only on the few pixels that actually contain the number instead of the whole noisy frame.

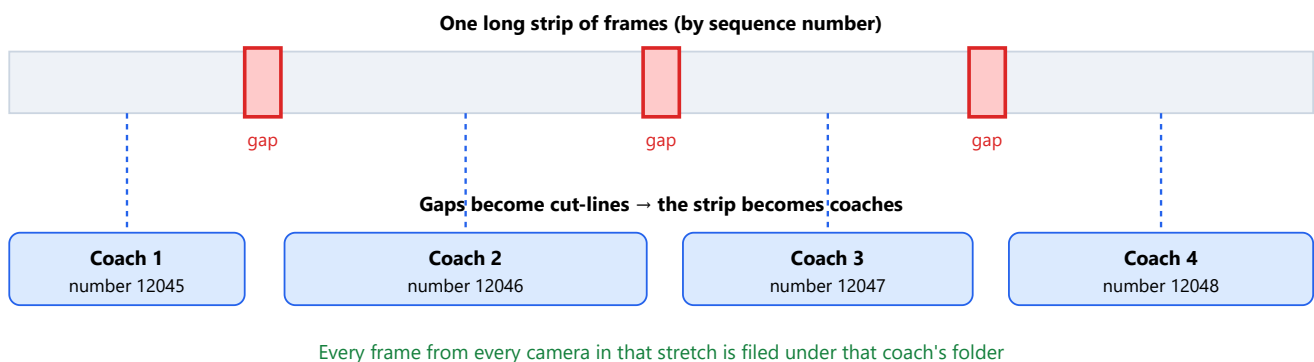
4. The Bogie Image Allocation System — Sorting Frames into Coaches

After splitting the video we have thousands of loose frames in one long strip — but no labels saying which frame belongs to which coach. The **Bogie Image Allocation System** (the Sync Engine) solves this: it cuts the long strip at the right places and files every frame under the correct coach.

Analogy: a long train of railway carriages photographed continuously. We need to know where one carriage ends and the next begins, then put each photo into the correct carriage's folder.

The clever clue: the gaps between coaches

There is always a visible **gap** where two coaches couple together. The YOLO Vision model is trained to spot that gap. Each gap is a natural "cut line" — a boundary between one bogie and the next.



Gaps between coaches act as exact cut-lines that divide the frame strip into individual coaches.

How the allocation actually happens

- **Collect the gap sightings.** As the train rolls, the same gap is seen across several frames. The system gathers all these "gap" detections in order.
- **Merge duplicates.** Nearby sightings of the *same* physical gap are clustered together so one gap counts once (it keeps the clearest sighting as the official boundary).
- **Cut the timeline.** Each boundary divides the sequence into stretches — one stretch per bogie/coach.
- **Name each coach.** Within each stretch it picks the best coach number that the OCR Reader found there.
- **File every frame.** Every frame — from *all* cameras — whose sequence number falls inside a stretch is assigned to that coach. Because gaps are physically exact, the matching is precise.

Safety net: if the gaps can't be seen in a particular run, the system falls back to grouping frames by voting on the coach numbers the OCR read — so it still works even when the gap clue is missing.

End result: a tidy hierarchy the operator can browse — **Train → Coach → Camera → Frame → Defect** — with every defect tied to a real, OCR-verified coach number.